Host Communication Documentation

SECS/GEM

- Chapter 1 -

General Requirements
State Models
Gem Compliance Statement
Gem Scenario
SECS-II Message
Message Detail
Data Item Dictionary

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0. Documentation History

Revision	Date	History	Author
0.0	07/07/06	Chapter1-3 for Y-Series	
1.0	10/07/02	Chapter4-9 for PMC	
1.1	10/08/10	Add S9F13, <MEXP> & <EDID>.	
1.2	10/10/25	Delete S10F5, S10F6, S10F7.	
1.3	10/12/03	Separate Chapter 1 & 2	
1.4	11/05/25	ALID format was changed. “52”->“52, 54”	
1.5	11/11/11	Add Format Code.	
1.6	15/11/11	Add the effective condition of spool Add the specific S/F used by S2F43	
1.7	19/10/03	YPM ALID format was changed. “52”->“52, 54”	
1.8	19/11/28	S1F15 OFF-LINE Event nothing	

1. General Requirements

This Host Communication System is based on below Standards.

	Title	Version	Source
HSMS	High-Speed SECS Message Services(HSMS) Generic Services.	SEMI E37-95	SEMI
SECS-II	SEMI Equipment Communications Standard 2, Message Content(SECS-II)	SEMI E5-0698	SEMI
GEM	SEMI Generic Model for Communications and Control of SEMI Equipment(GEM)	SEMI E30-0298	SEMI

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2. State Models

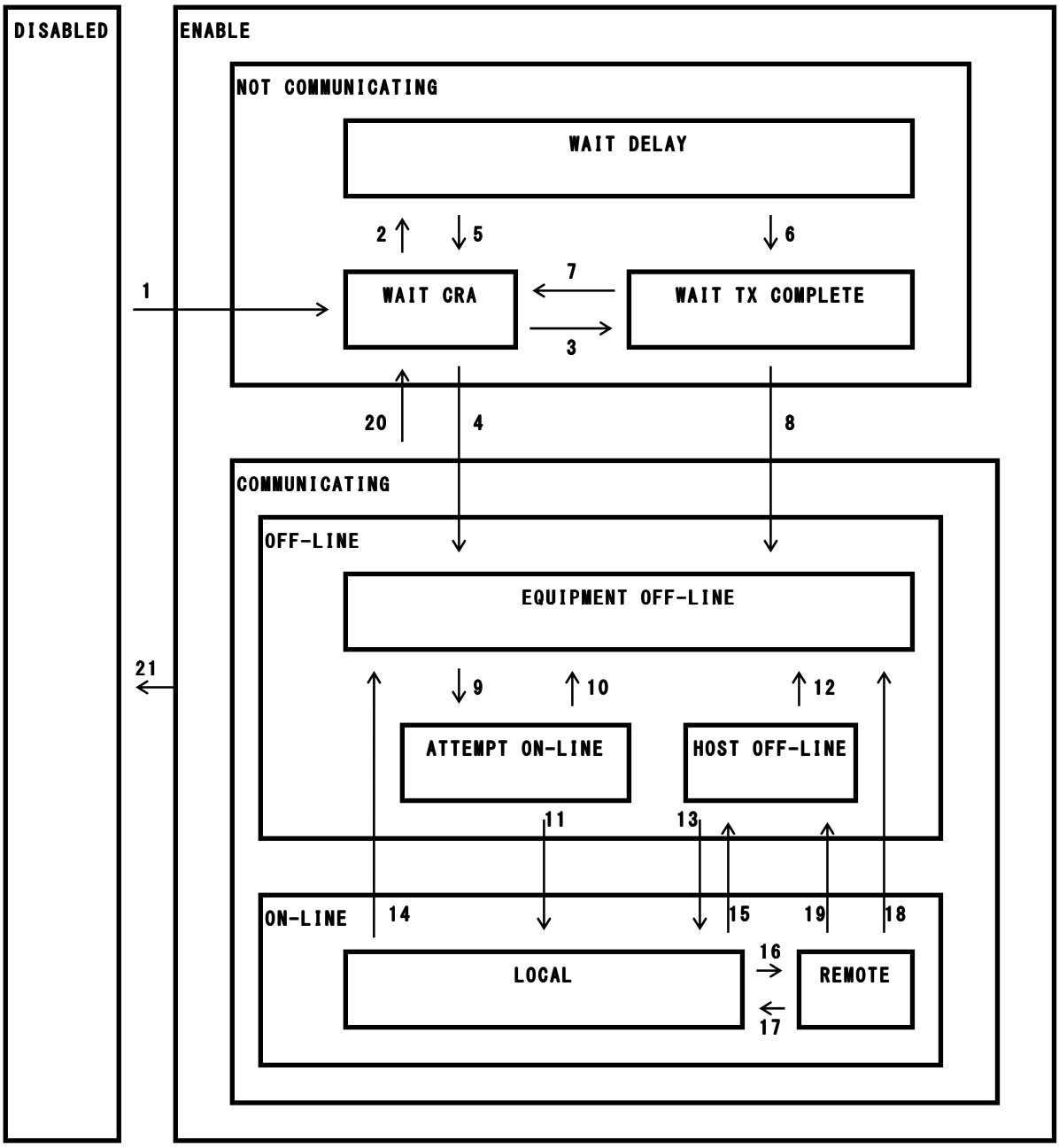
2-1 Communications and Control State Model

Communications and Control States Structure

DISABLED	: SECS-II Communication with a host computer is non-existent.			
ENABLED	: SECS-II Communication with a host computer is existent.			
	NOT COMMUNICATING	: The equipment is waiting Establish Communications.		
		WAIT CRA	: Waiting an Establish Communications Acknowledge.	
		WAIT DELAY	: Waiting between attempts to Establish Communications.	
		WAIT TX COMPLETE	: Wait for the successful completion of the S1F14.	
	COMMUNICATING	: Communications have been established.		
		OFF-LINE	: Impossible to transfer valid message.	
			EQUIPMENT OFF-LINE	: Waiting operator instructions to attempt.
			ATTEMPT ON-LINE	: Attempts to send an S1F1 to the host.
			HOST OFF-LINE	: The equipment is ON-LINE, but the host has not agreed.
		ON-LINE	: SECS-II message may be exchanged and acted upon.	
			LOCAL	: The equipment is operated only by operator.
			REMOTE	: The host may operate the equipment.

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Communications and Control State Diagram



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Communications and Control State Transition Table

Current State	#	Trigger	New State
DISABLED	1	System initialization	WAIT CRA
WAIT CRA	2	Connection transaction failure.	WAIT DELAY
	3	Received S1F13.	WAIT TX COMPLETE
	4	Received expected S1F14 with COMMACK = 0.	EQUIPMENT OFF-LINE
WAIT DELAY	5	CommDelay timer expired.	WAIT CRA
	6	Received S1F13.	WAIT TX COMPLETE
WAIT TX COMPLETE	7	Communication failure.	WAIT CRA
	8	S1F14 transmission completed Successfully.	EQUIPMENT OFF-LINE
EQUIPMENT OFF-LINE	9	Operator switches to ON-LINE.	ATTEMPT ON-LINE
ATTEMPT ON-LINE	10	Communication failure. Reply timeout.	EQUIPMENT OFF-LINE
	11	Received expected S1F2.	LOCAL
HOST OFF-LINE	12	Operator switches to OFF-LINE.	EQUIPMENT OFF-LINE
	13	Equipment accepts host request to go ON-LINE(S1F17)	LOCAL
LOCAL	14	Operator switches to OFF-LINE.	EQUIPMENT OFF-LINE
	15	Equipment accepts "Set OFF-LINE" message from host(S1F15)	HOST OFF-LINE
	16	Operator switches to REMOTE.	REMOTE
REMOTE	17	Operator switches to LOCAL.	LOCAL
	18	Operator switches to OFF-LINE.	EQUIPMENT OFF-LINE
	19	Equipment accepts "Set OFF-LINE" message from host(S1F15)	HOST OFF-LINE
COMMUNICATING	20	Communication failure.	WAIT CRA
ENABLED	21	Operator switches from DISABLED to DISABLED.	DISABLED

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2-2 Equipment Processing State Model

Equipment Processing State Definitions

1. DOWN

In this state the equipment can't move. It contains EMERGENCY STOP, KWY-SW OFF, SYSTEM READY OFF.

2. ACTIVE

This state is the parent of all substates where the context of process program execution exists.

3. READY

In this state the equipment is ready for process execution and is waiting a START command from the operator or the HOST.

4. AUTO

In this state the equipment is able to process the L/F in LINE AUTO mode and PROCESS AUTO mode.

5. MAINTE

In this state the operator maintain the equipment condition. The state contains CHASE CHANGE mode, CLEANING mode, and so on.

6. IDLE

In this state the equipment is able to start a process. This state is under the conditions those are AUTO mode and no L/F is in the equipment.

7. RUN

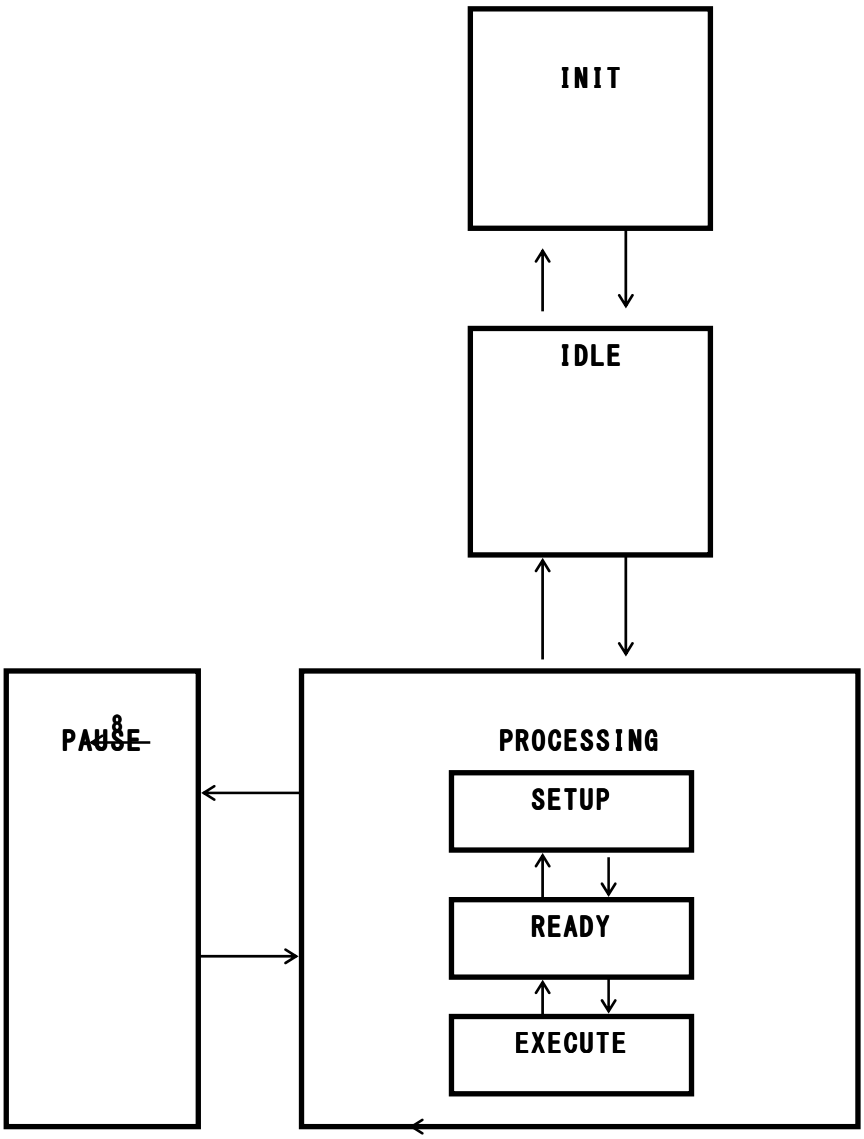
Run is the state in which the L/F are in the equipment and the equipment is processing the L/F automatically.

Equipment Processing States Structure

DOWN			: The equipment is impossible to move.
ACTIVE			: The equipment is possible move.
	READY		: Waiting a operation by an operator or a host.
	AUTO		: The equipment is possible to process execution automatically.
		IDLE	: The equipment is possible to start and waiting L/F.
		RUN	: The equipment is executing the L/F automatically.
	MAINTE		: Out of normal operation in order to maintain the equipment.

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Equipment Processing State Diagram



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Equipment Processing State Transition Table

Current State	#	Trigger	New State
DOWN	1	Turn on the switch to READY ON	READY
READY	2	Turn on the switch to AUTO	IDLE
	3	Turn on the switch to MAINTE	MAINTE
IDLE	4	Materials is supplies to the equipment	RUN
RUN	5	Complete the process	IDLE
AUTO	6	Turn off the switch from AUTO	READY
MAINTE	7	Turn off the switch from MAINTE	READY
ACTIVE	8	Turn off the switch from READY ON	DOWN

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3. Gem Compliance Statement

GEM COMPLIANCE STATEMENT		
FUNDAMENTAL GEM QUIREMENTS	**IMPLEMENTED**	**GEM COMPLIANT**
State Models	**Yes**	**Yes**
Equipment Processing States	**Yes**	
Host-Initiated S1, F13/F14 Scenario	**Yes**	
Event Notification	**Yes**	
On-Line Identification	**Yes**	
Error Messages	**Yes**	
Documentation	**Yes**	
Control(Operator Initiated)	**Yes**	
ADDITIONAL CAPABILITIES	**IMPLEMENTED**	**GEM COMPLIANT**
Establish Communications	**Yes**	**Yes**
Dynamic Event Report Configuration	**Yes**	**Yes**
Variable Data Collection	**Yes**	**Yes**
Trace Data Collection	**Yes**	**Yes**
Status Data Collection	**Yes**	**Yes**
Alarm Management	**Yes**	**Yes**
Remote Control	**Yes**	**Yes**
Equipment Constants	**Yes**	**No**
Process Program Management	**Yes**	**Yes**
Material Movement	**No**	**No**
Equipment Terminal Services	**Yes**	**Yes**
Clock	**Yes**	**Yes**
Limits Monitoring	**Yes**	**Yes**
Spooling	**Yes**	**Yes**
Control(Host-Initiated)	**Yes**	**Yes**

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4. Gem Scenario

1. Establish Communication

   1.1 Host Attempts To Establish Communications
   1.2 Equipment Attempts To Establish Communications and Host Acknowledges
   1.3 Equipment Receives S1F14 From Host Before Sending S1F14
   1.4 Equipment Sends S1F14 To Host Before Receiving S1F14

2. Event Notification

   2.1 Collection Event Occurs on the Equipment
   2.2 Host Requests Event Report

3. Dynamic Event Report Configuration

   3.1 Collection Event Reporting Set-up

4. Variable Data Collection

   4.1 Host Requests Report

5. Trace Data Collection

   5.1 Host Initiates Trace Report

6. Limits Monitoring

   6.1 Zone Transition Event occurs in Equipment:
   6.2 Host defines Limit Attributes:
   6.3 Host queries equipment for current Limits:

7. Status Data Collection

   7.1 Request Equipment Status Report
   7.2 Request Equipment Status Variable Namelist

8. On-line Identification

   8.1 Host Initiated
   8.2 Loopback Diagnostic

9. Alarm Management

   9.1 Enable/disable Alarms:
   9.2 Upload Alarm Information:
   9.3 Send Alarm Report:

10. Remote Control

   10.1 Remote Control

11. Equipment Constants

   11.1 Host Sends Equipment Constants
   11.2 Host Equipment Constants Request:
   11.3 Host Equipment Constant Namelist Request:

12. Process Program Management

   12.1 Process Program Creation, Editing, or Deletion Process Program Created, Edited, or Deleted by Operator:
   12.2 Process Program Deletion by the Host:
   12.3 Process Program Directory Request
   12.4 Host-Initiated Process Program Upload - Unformatted:
   12.5 Host-Initiated Process Program Download - Unformatted:
   12.6 Equipment-Initiated Process Program Upload - Unformatted:
   12.7 Equipment-Initiated Process Program Download - Unformatted:

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13.Equipment Terminal Services

13.1 Host Sends Information to an Equipment's Display device:
13.2 Host Sends Information to an Equipment's Display device and then overwrites the Information before Operator recognizes message:
13.3 Operator sends information to the host:
13.4 Host Send a Multi-Block Display Message

14.Error Messages

14.1 Message Fault Due to Unrecognized Device ID:
14.2 Message Fault Due to Unrecognized Stream Type:
14.3 Message Fault Due to Unrecognized Function Type:
14.4 Message Fault Due to Illegal Data Format:
14.5 Communication Fault Due to Transaction Timer Timeout:
14.6 Message Fault Due to data too long:

15.Clock

15.1 Equipment Request Time (Optional Scenario)
15.2 Host Instruct Equipment to Set Time.
15.3 Host Requests Equipment's Current Time Value:

16.Spooling

16.1 Define the Set of Messages to be Spooled:
16.2 Defines the Maximum Number of Messages to Send in Response to S6F23:
16.3 Request or Delete Spooled Data(MaxSpoolTransmit = 0):
16.4 Request or Delete Spooled Data(MaxSpoolTransmit > 0):

17.Control(Operator-Initiated Scenarios)

17.1 Host Accepts ON-LINE:
17.2 Host Denies ON-LINE:
17.3 Operator Sets OFF-LINE:
17.4 Operator Sets REMOTE:
17.5 Operator Sets LOCAL:

18.Control(Host-Initiated scenarios)

18.1 Host Sets OFF-LINE:
18.2 Host Sets ON-LINE:

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4-1. Establish Communications

4-1-1 Host Attempts To Establish Communications

COMMENT	HOST	EQUIPMENT	COMMENT
		Communications state is Enabled (any substate)	
Establish Communications Request	S1F13→		
		←S1F14 Reply COMMACK=Accept and Communications state = COMMUNICATING	

4-1-2 Equipment Attempts To Establish Communications and Host Acknowledges

COMMENT	HOST	EQUIPMENT	COMMENT
		Communications State = NOT COMMUNICATING	
		[LOOP]	
		[LOOP] --SEND	
		←S1F13 Establish Communications Request	
Establish Communications Acknowledge	S1F14→		
		[IF] S1F14 rcvd w/o timeouts	
		[THEN] exit_loop--SEND	
		[ELSE] Delay for interval in Establish Communications Timeout	
		[ENDIF]	
		[END_LOOP] --SEND	
		[IF] COMMACK=Accept	
		[THEN] Communications state=Communicating exit_loop--	
		[ELSE] Reset timer for delay, and delay for interval specified in EstablishCommunications Timeout	
		[ENDIF]	
		[END_LOOP]	

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4-1-3 Equipment Receives S1F14 From Host Before Sending S1F14

COMMENT	HOST	EQUIPMENT	COMMENT
		Communications State = NOT COMMUNICATING	
		←S1F13	Establish Communications Request
Establish Communications Request	S1F13→		
Reply COMMACK = Accept	S1F14→		S1F14 received from Host and Communications established and Communications state = COMMUNICATING
		←S1F14	Reply COMMACK=Accept

4-1-4 Equipment Sends S1F14 To Host Before Receiving S1F14

COMMENT	HOST	EQUIPMENT	COMMENT
		Communications State = NOT COMMUNICATING	
		←S1F13	Establish Communications Request
Establish Communications Request	S1F13→		
		←S1F14	Reply COMMACK=Accept Communications established and Communications state = COMMUNICATING
Reply COMMACK = Accept	S1F14→		S1F14 received from Host

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4-2. Event Notification

4-2-1 Collection Event Occurs on the Equipment

COMMENT	HOST	EQUIPMENT	COMMENT
			[IF]Event Report is Multi-block
		←S6F5	[THEN]send Multi-block inquire
Multi-block grant	S6F6 →		
			[ENDIF]
		←S6F11	Equipment sends Event Report
Host acknowledges Event Report	S6F12→		

4-2-2 Host Requests Event Report

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests an event report	S6F15→		
		←S6F16	Equipment sends event report

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4-3. Dynamic Event Report Configuration

4-3-1 Collection Event Reporting Set-up

COMMENT	HOST	EQUIPMENT	COMMENT
[IF]Define Report is Multi-block			
[THEN]send Multi-block inquire	S2F39→		
		←S2F40	Multi-block grant.
[ENDIF]			
Send report definitions	S2F33→		DATAIDs,RPTIDs and VIDs received.
		←S2F34	DRACK=0 the reports are OK
[IF]Link Event/Report if Multi-block			
[THEN]send Multi-block inquire	S2F39→		
		←S2F40	Multi-block grant.
[ENDIF]			
Link reports to events	S2F35→		CEIDs and the corresponding RPTIDs are received.
		←S2F36	LRACK=0 the event linkages are Acceptable
			Enable/disable codes(CEEDs) and the respective event reporting CEIDs received.
Enable specific collection events	S2F37→		
		←S2F38	ERACK=0 OK,will generate the specified reports when the appropriate collection events happen.

4-4. Variable Data Collection

4-4-1 Host Requests Report

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests data variables contained in report RPTID	S6F19→		
		←S6F20	Equipment responds with a list of variable data for the given RPTID.

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4-5. Trace Data Collection

4-5-1 Host Initiates Trace Report

COMMENT	HOST	EQUIPMENT	COMMENT
Trace Data initialization requested.	S2F23→		
		←S2F24	Acknowledge,trace initiated
			[DO]TOTSMP REPGSZ times
			[DO]REPGSZ many times:collect SVID1,...,SVIDn data, delay time by DSPER.
			[END_DO]
		←S6F1	Send SV1,...,SVn
Acknowledge receipt	S6F2 →		
			[END_DO]
Optional: Request trace termination prior to completion(TOTSMP=0)	S2F23→		
		←S2F24	Acknowledge premature termination

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4-6. Limits Monitoring

4-6-1 Zone Transition Event occurs in Equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
			[IF]Event Report is Multi-block
		←S6F5	[THEN]send Multi-block inquire
Multi-block grant	S6F6 →		
			[ENDIF]
		←S6F11	Equipment sends Event Report
Host acknowledges Event Report	S6F12→		

4-6-2 Host defines Limit Attributes:

COMMENT	HOST	EQUIPMENT	COMMENT
[IF]S2F45 is Musti-block			
[THEN]Send Multi-block inquire	S2F39→		
		←S2F40	Multi-block grant.
[END IF]			
Host defines new variable limit attributes.	S2F45→		
		←S2F46	Equipment acknowledges host request.

4-6-3 Host queries equipment for current Limits:

COMMENT	HOST	EQUIPMENT	COMMENT
Host queries equipment for current variable limit attribute definitions.	S2F47→		
		←S2F48	Equipment returns report containing requested variable limit attribute

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4-7. Status Data Collection

4-7-1 Request Equipment Status Report:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests report of selected status variable values.	S1F3→		
		←S1F4	Equipment responds with the requested status variable data.

4-7-2 Request Equipment Status Variable Namelist:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests equipment to identify selected status variables.	S1F11→		
		←S1F12	Equipment responds with the requested status variable descriptions.

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4-8. On-line Identification

4-8-1 Host Initiated:

COMMENT	HOST	EQUIPMENT	COMMENT
Are you there?	S1F1→		
		←S1F2	Equipment replies with MDLN and SOFTREV.

4-8-2 Loopback Diagnostic:

COMMENT	HOST	EQUIPMENT	COMMENT
Loopback Diagnostic Request	S2F25→		
		←S2F26	The binary string(ABS) sent is echoed back.

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4-9. Alarm Management

4-9-1 Enable/disable Alarms:

COMMENT	HOST	EQUIPMENT	COMMENT
Enable/disable Alarm	S5F3→		
		←S5F4	Acknowledge

4-9-2 Upload Alarm Information:

COMMENT	HOST	EQUIPMENT	COMMENT
Request alarm data/text	S5F5→		
		←S5F6	Send alarm data/text

4-9-3 Send Alarm Report:

COMMENT	HOST	EQUIPMENT	COMMENT
		Alarm occurrence detected by the equipment.	
		←S5F1	Send alarm report(if enabled).
Acknowledge	S5F2→		
		←S6F11	Send event report(if enabled)
Acknowledge	S6F12→		

4-10. Remote Control

4-10-1 Remote Control:

COMMENT	HOST	EQUIPMENT	COMMENT
Host Command Send	S2F41→		
		←S2F42	Host Command Acknowledge HCACK=0(assume command can be performed).
		←S6F11	Event Report-state change or other collection event occurrence.
Event Report Acknowledge	S6F12→		

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4-11. Equipment Constants

4-11-1 Host Sends Equipment Constants:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends equipment constants	S2F15→		
		←S2F16 EAC=0 equipment sets constants	

4-11-2 Host Equipment Constants Request:

COMMENT	HOST	EQUIPMENT	COMMENT
Host constant request	S2F13→		
		←S2F14 Equipment constant data	

4-11-3 Host Equipment Constant Namelist Request:

COMMENT	HOST	EQUIPMENT	COMMENT
Host constant namelist request	S2F29→		
		←S2F30 Equipment constant namelist	

4-11-4 Operator Changes Equipment Constant:

COMMENT	HOST	EQUIPMENT	COMMENT
		←S6F11 Operator changes equipment constant at equipment operator console. Equipment reports equipment constant change.	
Host acknowledges event	S6F12→		

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4-12. Process Program Management

4-12-1 Process Program Creation, Editing, or Deletion Process Program Created, Edited, or Deleted by Operator:

COMMENT	HOST	EQUIPMENT	COMMENT
			New process program created, edited, or deleted by operator at equipment. PPChangeName = PPID PPChangeStatus = 1 (Created) = 2 (Edited) = 3 (Deleted) [IF] CEID for Process Program Change Event enabled [THEN]
		←S6F11	Send Event Report
Event Report Acknowledge	S6F12→		

4-12-2 Process Program Deletion by the Host:

COMMENT	HOST	EQUIPMENT	COMMENT
Delete Process Program Send	S7F17→		
		←S7F18	Delete Process Program Acknowledge. The process program is removed from non-volatile storage.

4-12-3 Process Program Directory Request:

COMMENT	HOST	EQUIPMENT	COMMENT
Current EPPD Request	S7F19→		
		←S7F20	Current EPPD Data

4-12-4 Host-Initiated Process Program Upload - Unformatted:

COMMENT	HOST	EQUIPMENT	COMMENT
Process Program Request	S7F5 →		
		←S7F6	Process Program Data

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4-12-5 Host-Initiated Process Program Download - Unformatted:

COMMENT	HOST	EQUIPMENT	COMMENT
[IF]Process program is multi-block			
[THEN]			
Process Program Load Inquire	S7F1→		
		←S7F2	Process Program Load Grant
[END_IF]			
Process Program Send	S7F3→		
		←S7F4	Process Program Acknowledge

4-12-6 Equipment-Initiated Process Program Upload - Unformatted:

COMMENT	HOST	EQUIPMENT	COMMENT
			[IF]Process program is multi-block
			[THEN]
		←S7F1	Process Program Load Inquire
Process Program Load Grant	S7F2→		
			[END_IF]
		←S7F3	Process Program Send
Process Program Acknowledge	S7F4→		

4-12-7 Equipment-Initiated Process Program Download - Unformatted:

COMMENT	HOST	EQUIPMENT	COMMENT
		←S7F5	Process Program Request
Process Program Data	S7F6→		

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4-13. Equipment Terminal Services

4-13-1 Host sends information to an equipment's display device:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends textual information to equipment for display to the operator on terminal 0.	S10F3→		
		←S10F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.) Operator indecates message recognition. (Equipment clears unrecognized message indicator.)
		←S6F11	Message recognition event.
Host acknowledges.	S6F12→		
		←S10F1	Operator responds with text via terminal 0
Host acknowledges receipt of operator text	S10F2→		

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4-13-2 Host sends information to an equipment's display device and then overwrites the information before operator recognizes message:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends textual information to equipment for display to the operator on terminal 1.	S10F3→		
		←S10F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.)
Host sends textual information to equipment for display to the operator on terminal 1.This message overwrites the first one sent by the host since it is still unrecognized.	S10F3→		
		←S10F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.) Operator indicates message recognition. (Equipment clears unrecognized message indicator.)
		←S6F11	Message recognition event.
Host acknowledges.	S6F12→		

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4-13-3 Operator sends information to the host:

COMMENT	HOST	EQUIPMENT	COMMENT
		←S10F1	Operator sends textual information via equipment terminal x
Host acknowledges receipt of operator initiated message.	S10F2→		
Optional: Host responds with information for display to the operator on terminal x	S10F3→		
		←S10F4	Equipment acknowledges receipt of request to display text (Equipment sets unrecognized message indicator.) Operator indicates message recognition.
		←S6F11	Message recognition event.
Host acknowledges.	S6F12→		

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4-14. Equipment Terminal Services

4-14-1 Message Fault Due to Unrecognized Device ID:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects an unrecognized device ID within the message from the host.
		←S9F1	Equipment reports to the host that an "unrecognized device ID" was detected in the received message.

4-14-2 Message Fault Due to Unrecognized Stream Type:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects an unrecognized stream type within the message from the host.
		←S9F3	Equipment reports to the host that an "unrecognized stream type" was detected in the received message.

4-14-3 Message Fault Due to Unrecognized Function Type:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects an unrecognized function type within the message from the host.
		←S9F5	Equipment reports to the host that an "unrecognized function type" was detected in the received message.

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4-14-4 Message Fault Due to Illegal Data Format:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects illegal data format within the message from the host.
		←S9F7	Equipment reports to the host that "illegal data format" was detected in the received message.

4-14-5 Communication Fault Due to Transaction Timer Timeout:

COMMENT	HOST	EQUIPMENT	COMMENT
			Equipment does not receive an expected reply message from the host and a transaction timer timeout occurs.
		←S9F9	Equipment reports to the host that a transaction timer timeout occurred.

4-14-6 Message Fault Due to data too long:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	SxFy→		
			Equipment detects that the message from the host contains more data than it can handle.
		←S9F11	Equipment reports to the host that "data too long" was detected in the received message.

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4-14-7 Message Fault Due to Conversation Timeout:

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends a message.	S7F1→		
			Though data should have been received, Host was not able to receive it before appropriate time.
		←S9F13	Equipment reports to the host that a conversation timeout.

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4-15. Clock

4-15-1 Equipment Request Time (Optional Scenario):

COMMENT	HOST	EQUIPMENT	COMMENT
		←S2F17	Equipment requests a time value from the host
Host responds with a TIME value	S2F18→		
			Equipment sets its internal time reference to the value of TIME received from the host.

4-15-2 Host Instruct Equipment to Set Time:

COMMENT	HOST	EQUIPMENT	COMMENT
Host instructs equipment to set its time	S2F31→		
		←S2F32	Equipment sets its internal time reference to the value of TIME received from the host and acknowledges completion

---

4-15-3 Host Requests Equipment's Current Time Value:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests equipment time	S2F17→		
		←S2F18	Equipment returns its internal time reference value to the host.

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4-16. Spooling

The Spooling function is effective only when control status is “ON-LINE”.
The maximum size of spooled data is 1M byte.

4-16-1 Define the Set of Messages to be Spooled:

This Scenario is used to set up the list of messages that the equipment should spool(or by defining none, to disable spooling).
This setting is necessary before spooling start (before discommunicate).

COMMENT	HOST	EQUIPMENT	COMMENT
Host defines messages to be spooled in case of communications failure.	S2F43→		
		←S2F44	Equipment acknowledges setup.

4-16-2 Defines the Maximum Number of Messages to Send in Response to S6F23:

This Scenario sets the value of the equipment constant MaxSpoolTransmit.

COMMENT	HOST	EQUIPMENT	COMMENT
Host sends value for equipment constant MaxSpoolTransmit.	S2F15→		
		←S2F16	quipment acknowledges equipment constant change.

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4-16-3 Request or Delete Spooled Data(MaxSpoolTransmit = 0):

This Scenario is used to initiate the transfer of the spooled data from the equipment to the host or to purge the spool.

COMMENT	HOST	EQUIPMENT	COMMENT
		Communications were lost and then re-established	
Host requests data that includes spool related status variables.	S1F3→		
		←S1F4	Send status data
NOTE:S1F3 is one of various methods that could be used.	S6F23→		
Request or delete spooled data			
		←S6F24	Request spooled data acknowledgement. [IF]RSDC = 0(Spool data requested.) [THEN] The appropriate Streams and Functions are used to transmit the spooled data to the host. [ELSE_IF]RSDC = 1 [THEN] Spool data discarded. [END_IF]
		←S6F11	Spooling Deactivated event report sent.
Acknowledge	S6F12→		

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4-16-4 Request or Delete Spooled Data(MaxSpoolTransmit > 0):

This Scenario shows the affect of MaxSpoolTransmit<SpoolCountActual on the Spool Transmit process. For the purpose of illustration, the value of MaxSpoolTransmit is 5 and the SpoolCountActual is 8 (at the time communications are re-established). No messages are added to the Spool during the transmit process.

COMMENT	HOST	EQUIPMENT	COMMENT
Communications were lost and then re-established			
Host requests data that includes spool related status variables.	S1F3→		
		←S1F4	Send status data(e.g. SpoolCountActual = 8, MaxSpoolTransmit = 5).
Host requests spooled data(RSDC=0).	S6F23→		
		←S6F24	Request spooled data acknowledgement. The five oldest messages in the Spool are transmitted to the host. Spooling remains active.
Host recognizes that MaxSpoolTransmit is reached. Host requests additional spooled data (RSDC=0).	S6F23→		
		←S6F24	Request spooled data acknowledgement. The three remaining messages are transmitted from the spool.
		←S6F11	Spooling Deactivated event report sent.
Acknowledge	S6F12→		

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4-17. Control(Operator-Initiated Scenarios)

4-17-1 Host Accepts ON-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
Operator actuates ON-LINE switch when equipment OFF-LINE state is active.			
		←S1F1 Equipment requests ON-LINE.	
Host grants ON-LINE.	S1F2→		
		←S6F11 "Control State LOCAL(or REMOTE)"event.	
Acknowledge	S6F12→		

4-17-2 Host Denies ON-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
Operator actuates ON-LINE switch when equipment OFF-LINE state is active.			
		←S1F1 Equipment requests ON-LINE.	
Host denies ON-LINE.	S1F0→		

4-17-3 Operator Sets OFF-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
Operator actuates OFF-LINE switch when equipment ON-LINE state is active.			
		←S6F11 "Equipment requests OFF-LINE"event.	
Acknowledge.	S6F12→		

4-17-4 Operator Sets REMOTE:

COMMENT	HOST	EQUIPMENT	COMMENT
Operator sets switch from LOCAL to REMOTE.			
		←S6F11 "Control State REMOTE"event.	
Acknowledge.	S6F12→		

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4-17-5 Operator Sets LOCAL:

COMMENT	HOST	EQUIPMENT	COMMENT
		Operator sets switch from REMOTE to LOCAL.	
		←S6F11 "Control State LOCAL"event.	
Acknowledge.	S6F12→		

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4-18. Control(Host-Initiated scenarios)

4-18-1 Host Sets OFF-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests OFF-LINE.	S1F15→		
			[IF]Equipment is OFF-LINE
			[THEN]
		←S1F0	Equipment does not process requests.
			[ELSE]Equipment ON-LINE
		←S1F16	Equipment acknowledges request and transitions to OFF-LINE.
		~~←S6F11~~	~~"Equipment OFF-LINE"event.~~
Acknowledge	S6F12→		
			[END_IF]

4-18-2 Host Sets ON-LINE:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests ON-LINE.	S1F17→		
			[IF] Equipment is HOST OFF-LINE state not active.
			[THEN]
		←S1F18	Equipment denies request(ONLACK≠0).
			[ELSE] Equipment HOST OFF-LINE state is active.
		←S1F18	Equipment acknowledges Request (ONLACK=0).
		←S6F11	"Control stat LOCAL(or REMOTE)"event.
Acknowledge	S6F12→		
			[END_IF]

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5. SECS-II Message

5-1. Host Initiated Messages

HOST INITIATES			EQUIPMENT RESPONSE		
S/F	NAME	MN	S/F	NAME	MN
S1Fx	Stream 1 messages		S1F0	Abort Transaction	S1F0
S1F1	Are You There Request	R	S1F2	On Line Data	D
S1F3	Selected Equipment Status Request	SSR	S1F4	Selected Equipment Status Request	SSD
S1F11	Status Variable Namelist Request	SVNR	S1F12	Status Variable Namelist Reply	SVNRR
S1F13	Establish Communications Request	CR	S1F14	Establish Communications Request Acknowledge	CRA
S1F15	Request OFF-LINE	ROFL	S1F16	OFF-LINE Acknowledge	OFLA
S1F17	Request ON-LINE	RONL	S1F18	ON-LINE Acknowledge	ONLA
S2Fx	Stream 2 messages		S2F0	Abort Transaction	S2F0
S2F13	Equipment Constant Request	ECR	S2F14	Equipment Constant Data	ECD
S2F15	New Equipment Constant Send	ECS	S2F16	New Equipment Constant Acknowledge	ECA
S2F17	Data and Time Request	DTR	S2F18	Data and time Data	DTD
S2F23	Trace Initialize Send	TIS	S2F24	Trace Initialize Acknowledge	TIA
S2F25	Loopback Diagnostic Request	LDR	S2F26	Loopback Diagnostic Data	LDD
S2F29	Equipment Constant Namelist Request	ECNR	S2F30	Equipment Constant Namelist	ECN
S2F31	Date and Time Set Request	DTS	S2F32	Date and Time Set Acknowledge	DTA
S2F33	Define Report	DR	S2F34	Define Report Acknowledge	DRA
S2F35	Link Event Report	LER	S2F36	Link Event Report Acknowledge	LERA
S2F37	Enable/Disable Event Report	EDER	S2F38	Enable/Disable Event Report Acknowledge	EERA
S2F39	Multi-block Inquire	DMBI	S2F40	Multi-block Grant	DMBG
S2F41	Host command Send	HCS	S2F42	Host Command Acknowledge	HCA
S2F43	Reset Spooling Streams and Functions	RSSF	S2F44	Reset Spooling Acknowledge	RSA
S2F45	Define Variable Limit Attributes	DVLA	S2F46	Variable Limit Attribute Acknowledge	VLAA
S2F47	Variable limit Attribute Request	VLAR	S2F48	Variable Limit Attributes Send	VLAS
S5Fx	Stream 5 messages		S5F0	Abort Transaction	S5F0
S5F3	Enable/Disable Alarm Send	EAS	S5F4	Enable/Disable Alarm Acknowledge	EAA
S5F5	List Alarms Request	LAR	S5F6	List Alarm Data	LAD
S6Fx	Stream 6 messages		S6F0	Abort Transaction	S6F0
S6F15	Event Report Request	ERR	S6F16	Event Report Data	ERD
S6F19	Individual Report Request	IRR	S6F20	Individual Report Data	IRD
S6F23	Request Spooled Data	RSD	S6F24	Request Spooled Data Acknowledgement Send	RSDAS
S7Fx	Stream 7 messages		S7F0	Abort Transaction	S7F0
S7F1	Process Program Load Inquire	PPI	S7F2	Process Program Load Grant	PPG
S7F3	Process Program Send	PPS	S7F4	Process Program Ack	PPA
S7F5	Process Program Request	PPR	S7F6	Process Program Data	PPD
S7F17	Delete Process Program Send	DPS	S7F18	Delete Process Program Acknowledge	DPA
S7F19	Current EPPD Request	RER	S7F20	Current EPPD Data	RED
S10Fx	Stream 10 messages		S10F0	Abort Transaction	S10F0
S10F3	Terminal Display, Single	VTN	S10F4	Terminal Display, Single Acknowledge	VTA

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5-2. Equipment Initiated Messages

EQUIPMENT INITIATES			HOST RESPONSE		
S/F	NAME	MN	S/F	NAME	MN
S1Fx	Stream 1 messages		S1F0	Abort Transaction	S1F0
S1F1	Are You There Request	R	S1F2	On Line Data	D
S1F13	Establish Communications Request	CR	S1F14	Establish Communications Request Acknowledge	CRA
S2Fx	Stream 2 message		S2F0	Abort Transaction	S2F0
S2F17	Data and Time Request	DTR	S2F18	Data and time Data	DTD
S5Fx	Stream 5 messages		S5F0	Abort Transaction	S5F0
S5F1	Alarm Report Send	ARS	S5F2	Alarm Report Acknowledge	ARA
S6Fx	Stream 6 messages		S6F0	Abort Transaction	S6F0
S6F1	Trace Data Send	TDS	S6F2	Trace Data Acknowledge	TDA
S6F5	Multi-Bloc Data Send Inquire	MBI	S6F6	Multi-Block Grant	MBG
S6F11	Event Report Send	ERS	S6F12	Event Report Acknowledge	ERA
S7Fx	Stream 7 messages		S7F0	Abort Transaction	S7F0
S7F1	Process Program Load Inquire	PPI	S7F2	Process Program Load Grant	PPG
S7F3	Process Program Send	PPS	S7F4	Process Program Ack	PPA
S7F5	Process Program Request	PPR	S7F6	Process Program Data	PPD
S9F1	Unrecognized R-bit and Device ID	UDN	NONE		
S9F3	Unrecognized W-bit and Stream Type	USN	NONE		
S9F5	Unrecognized Function Type	UFN	NONE		
S9F7	Illegal Data	IDN	NONE		
S9F9	Transaction Timer Timeout	TTN	NONE		
S9F11	Data Too Long	DLN	NONE		
S9F13	Conversation Timeout	CTN	NONE		
S10Fx	Stream 10 messages		S10F0	Abort Transaction	S10F0
S10F1	Terminal Request	TRN	S10F2	Terminal Request Acknowledge	TRA

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6. Message Detail

6-1. Stream1 Equipment Status

S1F0 Abort Transaction(S1F0) S,H<->E

Description: Used is lieu of an expected replay to abort a transaction. Function 0 is defined in every stream and has the same meaning in every stream.

Structure: Header only

S1F1 Are You There Request(R) S,H<->E,reply

Description: Establishes if the equipment is on line. A function 0 response to this message means the communication is inoperative. In the equipment, a function 0 is equipment to a timeout on the receive timer after issuing S1F1 to the host.

Structure: Header only

S1F2 On Line Data(D) S,H<->E

Description: Data signifying that the equipment is alive.

Structure: L,2
1.<MDLN>
Equipment Model Type, 6bytes max
2.<SOFTREV>
Software revision code 6bytes max

Exception: The host sends a zero-length list to the equipment.

S1F3 Selected Equipment Status Request(SSR) S,H->E,reply

Description: A request to the equipment to report selected values of its status.

Structure: The following structure is approved for all item formats and should be used by all new implementations.

L,n
1.<SVID1>
Status variable ID
.
.
n.<SVIDn>

Exception: A zero-length list means report all SVIDs.

S1F4 Selected Equipment Status Data(SSD) M,H<-E

Description: The equipment reports the value of each SVID requested in the order requested. The host remembers the names of values requested.

Structure: L,n
1.<SV1>
Status variable value
.
.
n.<SVn>

Exception: If n=0, no response can be made. A zero length returned for SVi means that SVIDi does not exist.

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S1F11 Status Variable Namelist Request(SVNR) S,H->E,reply

Description: A request to the equipment to identify certain status variables.

Structure:
```
L,n
  1.<SVID1>
    Status variable ID
  .
  .
  n.<SVIDn>
```

Exception: A zero length means report all SVIDs.

S1F12 Status Variable Namelist Reply(SVNRR) M,H<-E

Description: The equipment reports to the host the name and units of the requested SVs.

Structure:
```
L,n
  1.L,3
    1.<SVID1>
      Status variable ID
    2.<SVNAME1>
      Status Variable Name 30 characters
    3.<UNITS1>
      Units Identifier, 6bytes
  2.L,3
  .
  .
  n.L,3
    1.<SVIDn>
    2.<SVNAMEn>
    3.<UNITSn>
```

S1F13 Establish Communications Request(CR) S,H<->E,reply

Description: The purpose of this message is to provide a formal means of initializing communications at a logical application level both on power-up and following a break in communications. It should be the first message sent following any period where host and Equipment SECS applications are unable to communicate. An attempt to send an Establish Communications Request(S1F13) should be repeated at programmable intervals until an Establish Communications Acknowledge(S1F14) is received within the transaction timeout period with an acknowledgement code accepting the establishment.

Structure:
```
L,2
  1.<MDLN>
    Equipment Model Type, 6bytes max
  2.<SOFTREV>
    Software revision code 6bytes max
```

Exception: The host sends a zero-length list to the equipment.

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S1F14 Establish Communications Request Acknowledge(CRA) S,H<->E

Description: Accept or deny Establish Communications Request(S1F13).MDLN and SOFTREV are on-line data and are valid only if COMMAK = 0.

Structure:
```
L,2
  1. <COMMACK>
    Establish Communications Acknowledge Code, 1byte
      0 = Accepted
      1 = Denied, Try Again
      65 = System Error
  2. L,2
    1.<MDLN>
      Equipment Model Type, 6bytes max
    2.<SOFTREV>
      Software revision code 6bytes max
```

Exception: The host sends a zero-length list to the equipment.

S1F15 Request OFF-LINE(ROFL) S,H->E,reply

Description: The host requests that the equipment transition to the OFF-LINE state.

Structure: Header only.

S1F16 OFF-LINE Acknowledge(OFLA) S,H<-E

Description: Acknowledge or error.

Structure:
```
<OFLACK>.
Acknowledge code for OFF-LINE request.
  0 = OFF-LINE Acknowledge
  65 = System Error
```

S1F17 Request ON-LINE(RONL) S,H->E,reply

Description: The host requests that the equipment transition to the ON-LINE state.

Structure: Header only.

S1F18 ON-LINE Acknowledge(ONLA) S,H<-E

Description: Acknowledge or error.

Structure:
```
<ONLACK>.
Acknowledge code for ON-LINE request.
  0 = ON-LINE Accepted
  65 = System Error
```

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6-2. Stream2 Equipment Control and Diagnostics

S2F0 Abort Transaction(S2F0) S,H<->E

Description: Same from as S1F0

S2F13 Equipment Constant Request(ECR) S,H ->E,reply

Description: Constants such as for calibration, servo gain, alarm limits, data collection mode and other values that are changed infrequently can be obtained using this message.

Structure: The following structure is approved for all item formats and should be used by all new implementations:

```
L,n
  1.<ECID1>
    Equipment Constant ID
  .
  .
  n.<ECIDn>
```

Exception: A zero-length list means report all ECV's according to a predefined order.

S2F14 Equipment Constant Data(ECD) M,H<-E

Description: Data Response to S2F13 in the order requested.

Structure: The following structure is approved for all item formats and should be used by all new implementations:

```
L,n
  1.<ECV1>
    Equipment Constant Value
  .
  .
  n.<ECVn>
```

Exception: If n=0, then no response exists. A zero length ECVi means that ECIDi does not exist.

S2F15 New Equipment Constant Send(ECS) S,H->E,reply

Description: Change one or more equipment constants.

Structure:

```
L,n
  1.L,2
    1.<ECID1>
      Equipment Constant ID
    2.<ECV1>
      Equipment Constant Value
  2.L,2
  .
  .
  n.L,2
    1.<ECIDn>
    2.<ECVn>
```

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S2F16 New Equipment Constant Acknowledge(ECA) S,H<-E

Description: Acknowledge or error. If EAC contains a non-zero error code, the equipment should not change any of the ECIDs specified in S2F15.

Structure:
<EAC>
Equipment acknowledge code, 1byte
0 = Acknowledge
1 = Defined, At least one constant does not exis
2 = Defined, Busy
3 = Defined, At least one constant out of range
65 = System Error

S2F17 Date and Time request(DTR) S,H<->E,reply

Description: Useful to check equipment time base or for equipment to synchronize with the host time base.

Structure: Header only

S2F18 Date and Time data(DTD) S,H<->E

Description: Actual time data.

Structure:
<TIME>
Time of day, 16bytes, However this item could be 12 bytes or 14 bytes.
If 16bytes, the format is YYYYMMDDhhmmsscc,
If 14bytes, the format is YYYYMMDDhhmmss,
If 12bytes, the format is YYMMDDhhmmss.

YYYY	=	Year	0000	to	9999
MM	=	Month	01	to	12
DD	=	Day	01	to	31
hh	=	Hour	00	to	59
mm	=	Minute	00	to	59
ss	=	Second	00	to	59
cc	=	Centisecond	00	to	99

Note: This item format is associated ECV(Time Format) which is supported as a configurable variable.
Refer to ECV, Time Format.
This is a format requirement only and does not imply either precision.

Exception: A zero length item means no time exists.

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S2F23 Trace Initialize Send(TIS) S,H->E,reply

Description: Status variables exist at all times. This function provides a way to sample a subset of those status variables as a function of time. The trace data is returned on S6F1 and is related to the original request by the TRID. Maximum 8 trace requests may be made to that equipment allowing it. If equipment receives S2F23 with the same TRID as a trace function that is currently in progress, the equipment should terminate the old trace and then initiate the new trace. A trace function currently in progress may be terminated by S2F23 with TRID of that trace and TOTSMP = 0.

Structure: The following structure is approved for all item formats and should be used by all new implementations:

```
L,2
  1.<TRID>
    Trace request ID
  2.<DSPER>
    Data sample period hhmmss, 6bytes
  3.<TOTSMP>
    Total samples to be made
  4.<REPGSZ>
    Reporting group size
  5.L,n
    1,<SVID1>
      Status variable ID
    .
    .
    n<SVIDn>
```

S2F24 Trace Initialize Acknowledge(TIA) S,H<-E

Description: Acknowledge or error.

Structure:

```
<TIAACK>
  Equipment acknowledgement code, 1 byte
    0 = everything correct
    1 = too many SVIDs
    2 = no more traces allowed
    3 = invalid period
    65 = System Error
```

S2F25 Loopback Diagnostic Request(LDR) S,H<->E,reply

Description: A diagnostic message for checkout of protocol and communication circuits. The binary string sent is echoed back.

Structure:

```
<ABS>
Any binary string
```

S2F26 Loopback Diagnostic Data(LDD) S,H<->E

Description: The echoed binary string.

Structure:

```
<ABS>
Any binary string
```

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S2F29 Equipment Constant Namelist Request(ECNR) S,H->E,reply

Description: This function allows the host to retrieve basic information about what equipment constants are available in the equipment.

Structure:
```
L,n
  1.<ECID1>
    Equipment Constant ID
  .
  .
  n.<ECIDn>
```

Exception: A zero length list means send information for all ECIDs.

S2F30 Equipment Constant Namelist(ECN) M,H<-E

Description: Data Response.

Structure:
```
L,n(number of equipment constants)
  1.L,6
    1.<ECID1>
      Equipment Constant ID
    2.<ECNAME1>
      Equipment constant name limited to 32 characters
    3.<ECMIN1>
      Equipment constant minimum value
    4.<ECMAX1>
      Equipment constant maximum value
    5.<ECDEF1>
      Equipment constant default value
    6.<UNITS1>
      Units Identifier, 6bytes
  .
  .
  .
  n.L,6
    1.<ECIDn>
    2.<ECNAMEn>
    3.<ECMINn>
    4.<ECMAXn>
    5.<ECDEFn>
    6.<UNITSn>
```

S2F31 Date and Time Set Request(DTS) S,H->E,reply

Description: Useful to synchronize the equipment time with the host time base.

Structure:
```
<TIME>
Time of day, 16bytes, However this item could be 12bytes or 14bytes.
  If 16bytes, the format is YYYYMMDDhhmmsscc,
  If 14bytes, the format is YYYYMMDDhhmmss,
  If 12bytes, the format is YYMMDDhhmmss.
     YYYY = Year          0000  to  9999
     MM   = Month         01    to  12
     DD   = Day           01    to  31
     hh   = Hour          00    to  59
     mm   = Minute        00    to  59
     ss   = Second        00    to  59
     cc   = Centisecond   00    to  99
```

Note: This item format is associated ECV(Time Format) which is Supported as a configurable variable.
Refer to ECV, Time Format.
This is a format requirement only and does not imply either precision.

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S2F32 Date and Time Set Acknowledge(DTA) S,H<-E

Description: Acknowledge the receipt of time and date.

Structure:
```
<TIACK>
Time acknowledge code, 1byte
  0 = OK
  1 = Error, not done
  65 = System Error
```

S2F33 Define Report(DR) M,H->E,reply

Description: The purpose of this message is for the host to define a group of reports for the equipment. If S2F33 is Multi-block, it must be preceded by the S2F39/S2F40 Inquire/Grant transaction.

Structure:
```
L,2
  1.<DATAID>
    Data ID
  2.L,a
    1.L,2
      1.<RPTID1>
        Report ID
      2.L,b
        1.<VID1>
          Variable ID
        .
        .
        b.<VIDb>
    .
    .
    a.L,2
      1.<RPTIDa>
      2.L,c
        1.<VID1>
        .
        .
        c.<VIDc>
```

Exception:
1. A list of zero length following <DATAID> deletes all report definitions and associated links. See S2F35(Link Event/Report).
2. A list of zero length following <RPTID> deletes report type RPTID. All CEID links to this RPTID are also deleted.

S2F34 Define Report Acknowledge(DRA) S,H<-E

Description: Acknowledge or error. If an error condition is detected the entire message is rejected, i.e. partial changes are not allowed.

Structure:
```
<DRACK>
Define Report Acknowledge Code, 1 byte
  0 = Accepted
  1 = Denied. Insufficient space
  2 = Denied. Invalid format
  3 = Denied. At least one RPTID already defined
  4 = Denied. At least one VID does not exist
  65 = System Error
```

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S2F35 Link Event Report(LER) M,H->E,reply

Description: The purpose of this message is for the host to link n reports to an event(CEID). These linked event reports will default to 'disabled' upon linking. That is, the occurrence of an event would not cause the report to be sent until enabled. See S2F37 for enabling reports.

Structure:
```
L,2
  1.<DATAID>
    Data ID
  2.L,a
    1.L,2
      1.<CEID1>
        Collected event ID
      2.L,b
        1.<RPTID1>
          Report ID
        .
        .
          b.<RPTIDb>
    .
    .
    a.L,2
      1.<CEIDa>
      2.L,c
        1.<RPTID1>
        .
        .
        c.<RPTIDc>
```

Exception: A list of zero length following CEID deletes all report links to that event.

S2F36 Link Event Report Acknowledge(LERA) S,H<-E

Description: Acknowledge or error. If an error condition is detected the entire message is rejected, i.e. partial changes are not allowed.

Structure:
```
<LRACK>
Link Report Acknowledge Code,1byte
  0 = Accepted
  1 = Denied. Insufficient space
  2 = Denied. Invalid format
  3 = Denied. At least one CEID link already defined
  4 = Denied. At least one CEID does not exist
  5 = Denied. At least one RPTID does not exist
  65 = System Error
```

S2F37 Enable/Disable Event Report(EDER) S,H->E,reply

Description: The purpose of this message is for the host to enable or disable reporting for a group of events (CEIDs)

Structure:
```
L,2
  1.<CEED>
    Collection event enable/disable code, 1 byte
      FALSE = Disable
      TRUE = Enable
  2.L,n
    1.<CEID1>
      Collected event ID
    .
    .
    n.<CEIDn>
```

Exception: A list of zero length following <CEED> means all CEIDs.

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S2F38 Enable/Disable Event Report Acknowledge(EERA) S,H<-E

Description: Acknowledge or error. If an error condition is detected the entire message is rejected, i.e. partial changes are not allowed.

Structure:
<ERACK>
Enable/Disable Event Report Acknowledge Code, 1 byte
0 = Accepted
1 = Denied. At least one CEID does not exist
65 = System Error

S2F39 Multi-block Inquire (DMBI) S,H->E,reply

Description: If a S2F33,S2F35 or S2F45 message is more than one block, this transaction must precede the message.

Structure:
L,2
1.<DATAID>
Data ID
2.<DATALENGTH>
Total bytes to be sent

S2F40 Multi-block Grant(DMBG) S,H<-E

Description: Grant permission to send multi-block message.

Structure:
<GRANT>
Grant code, 1 byte
0 = positive response, load OK
1 = busy, try again
2 = no space
3 = duplicate name
65 = System Error

S2F41 Host Command Send(HCS) S,H->E,reply

Description: The Host requests the Equipment perform the specified remote command with the associated parameters.

Structure:
L,2
1.<RCMD>
Remote command code or string
2.L,n# of parameters
1.L,2
1.<CPNAME1> parameter 1 name
Command Parameter Name
2.<CPVAL1> parameter 1 value
Command Parameter Value
.
.
n.L,2
1.<CPNAMEn> parameter n name
2.<CPVALn> parameter n value

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S2F42 Host Command Acknowledge(HCA) S,H<-E

Description: Acknowledge Host command or error. If command is not accepted due to one or more invalid parameters. (i.e. HCACK = 3), then a list of invalid parameters will be returned containing the parameter name and reason for being invalid.

Structure:
```
L,2
  1.<HCACK>
    Host Command Parameter Acknowledge Code, 1byte
      0 =      Acknowledge, command has been performed
      1 =      Command does not exist
      2 =      Cannot perform now
      3 =      At least one parameter is invalid
      4 =      Acknowledge, command will be performed with completion signaled
               later by an event.
      5 - 63 = Reserved
  2.L,n# of parameters
    1.L,2
      1.<CPNAME1> parameter 1 name
        Command Parameter Name
      2.<CPACK1> parameter 1 reason
        Command Parameter Value
    .
    .
    n.L,2
      1.<CPNAMEn> parameter n name
      2.<CPACKn>parameter n reason
```

Exception: If there are no invalid parameters then a list of zero length will be sent for item 2.

S2F43 Reset Spooling Streams and Functions(RSSF) S,H->E,reply

Description: This message allows the host to select specific streams and functions to be spooled whenever spooling is active.

Structure:
```
L,m
  1.L,2
    1.<STRID1>
      Stream Identification
    2.L,n
      1.<FCNID1>
        Function Identification
    .
    .
      n.<FCNIDn>
  .
  .
  m.L,2
    1.<STRIDm>
    2.L,n
      1.<FCNID1>
    .
    .
      n.<FCNIDn>
```

Exception:
1. A zero length list, m = 0, turns off spooling for all streams and functions.
2. A zero length list, n = 0, turns on spooling for all functions for the associated stream.

Note:
1. Turning off spooling for all functions for a specific stream is achieved by omitting reference to the stream from this message.
2. Spooling for Stream 1 is not allowed.
3. Equipment must allow host to spool all primary messages for a stream (except Stream 1).
   Specific:S2F17,S5F1,S6F1,S6F11,S7F3,S7F5,S10F1
4. A defined list of functions for a stream in this message will replace any previously selected functions.

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S2F44 Reset Spooling Acknowledge(RSA) M,H<-E

Description: Acknowledge or error.

Structure:
```
L,2
  1.<RSPACK>(accept or reject)
    Reset Spooling Acknowledge
      0 = Acknowledge, spooling setup accepted
      1 = Spooling setup rejected
  2.L,m (m = number of streams with errors)
    1.L,3
      1.<STRID1>
        Stream Identification
      2.<STRACK1>(error in stream)
        Spool Stream Acknowledge
          1 = Spooling not allowed for stream(i.g. Stream 1).
          2 = Stream unknown.
          3 = Unknown function specified for this stream.
          4 = Secondary function specified for this stream.
          Note: Error codes 1 - 4 shown in order of precedence for reporting.
Code 0 not used.
      3.L,n (n = number of functions in error)
        1.<FCNID1>
          Function Identification
        .
        .
        n.<FCNIDn>
      .
      .
    m.L,3
      1.<STRIDm>
      2.<STRACKm>(error in stream)
      3.L,n(n = number of functions in error)
        1.<FCNID1>
        .
        .
        n.<FCNIDn>
```

Exception:
1. If RSACK = 0, a zero length list, m = 0, is given, indicating no streams or functions in error.
2. A zero length list, n = 0, indicates no function in error for specified stream.

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S2F45 Define Variable Limit Attributes(DVLA) M,H->E,reply

Structure:

```
L,2
  1.<DATAID>
    Data ID
  2.L,m (m = number of variables in this definition)
    1.L,2
      1.<VID1>
        Variable ID
      2.L,n(n = number of limits being defined/changed for VID1)
        1.L,2
          1.<LIMITID1>
            The identifier of a specific limit in the set of limits for a variable
          2.L,p(p={0,2})
            1.<UPPERDB1>
              The variable limit attributes which defines the upper boundary of
              the deadband of a limit.
            2.<LOWERDB1>
              The variable limit attributes which defines the lower boundary of
              the deadband of a limit.
        .
        .
        n.L,2
          1.<LIMITIDn>
          2.L,p(p={0,2})
            1.<UPPERDB1>
            2.<LOWERDB1>
    .
    .
    m.L,2
      1.<VIDm>
      2.L,n(n = number of limits being defined/changed for VIDm)
        1.L,2
          1.<LIMITID1>
          2.L,p(p={0,2})
            1.<UPPERDB1>
            2.<LOWERDB1>
        .
        .
        n.L,2
          1.<LIMITIDn>
          2.L,p(p={0,2})
            1.<UPPERDB1>
            2.<LOWERDB1>
```

Exception:

1. A zero length list, m = 0, sets all limit values for all monitored VIDs to "undefined".
2. A zero length list, n = 0, sets all limits values for that VID to "undefined".
3. A zero length list, p = 0, sets that limit to "undefined".

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S2F46 Variable Limit Attribute Acknowledge(VLAA) M,H<-E

Description: Acknowledge definition of variable limit attributes or report error. If DVLA is not accepted due to one or more invalid parameters (e.g.LIMITACK=3), then a list of invalid parameters is returned containing the variable limit attribute and reason for rejection. If an error condition is detected, the entire message is rejected, i.e. partial changes are not allowed.

Structure:
```
L,2
  1.<VLAACK>
    Variable Limit Attribute Acknowledge Code, 1byte
      0 = Acknowledge, command will be performed
      1 = Limit attribute definition error
      2 = Cannot perform now
      65 = System Error
  2.L,m (m = number of invalid parameters)
    1.L,3
      1.<VID1> (VID with error)
        Variable ID
      2.<LVACK1> (reason)
        Variable limit definition acknowledges code, 1byte.
          1 = Variable does not exist.
          2 = Variable has no limits capability.
          3 = Variable repeated in message.
          4 = Limit value error as described in LIMITACK.
      3.L,n {n=0,2}
        1.<LIMITID1>(1st limit in error for VIDp)
          The identifier of a specific limit in the set of limits for a variable
        2.<LIMITACK1>(reasons)
          Acknowledgment code for variable limit attribute set, 1byte
          1 = LIMITID does not exist.
          2 = UPPERDB > LIMITMAX.
          3 = LOWERDB < LIMITMIN.
          4 = UPPERDB < LOWERDB.
          5 = Illegal format specified for UPPERDB or LOWERDB.
          6 = ASCII value limit definition for this variable.
      .
      .
      m.L,3
        1.<VIDm> (VID with error)
        2.<LVACKm> (reason)
        3.L,n {n=0,2}
          1.<LIMITIDm>(1st limit in error for VIDx)
          2.<LIMITACKm>(reasons)
```

Exception:
1. A zero length list, m = 0 indicates no invalid variable limit attributes.
2. A zero length list, n = 0 indicates no invalid limit values for that VID.

S2F47 Variable Limit Attribute Request(VLAR) S,H->E,reply

Description: This message allows the host to query the equipment for current variable limit attribute definitions.

Structure:
```
L,m
  1.<VID1>
    Variable ID
  .
  .
  m.<VIDm>
```

Exception: A zero length list, m = 0, requests a list of all VID values that can have variable limit attributes.

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S2F48 Variable Limit Attributes Send(VLAS) M,H<-E

Description: Equipment sends values of requested variable limit attribute definitions in the order requested.

Structure:

```
L,m
  1.L,2
    1.<VID1>
      Variable ID
    2.L,p {p=0,4}
      1.<UNITS1>
        Units Identifier, 6bytes
      2.<LIMITMIN1>
        The minimum allowed value for the limit values of a specific variable.
      3.<LIMITMAX1>
        The maximum allowed value for the limit values of a specific variable.
      4.L,n
        1.L,3
          1.<LIMITID1>
            The identifier of a specific limit in the set of limits for a variable
          2.<UPPERDB1>
            The variable limit attribute which defines the upper boundary of the
            deadband of a limit.
          3.<LOWERDB1>
            The variable limit attribute which defines the lower boundary of the
            deadband of a limit.
        .
        .
        n.L,3
          1.<LIMITIDn>
          2.<UPPERDBn>
          3.<LOWERDBn>
  .
  .
    m.L,2
      1.<VIDm>
      2.L,p {p=0,4}
        1.<UNITSm>
        2.<LIMITMINm>
        3.<LIMITMAXm>
        4.L,n
          1.L,3
            1.<LIMITID1>
            2.<UPPERDB1>
            3.<LOWERDB1>
        .
        .
        n.L,3
          1.<LIMITIDn>
          2.<UPPERDBn>
          3.<LOWERDBn>
```

Exception:

1. A zero length list, p = 0 indicates that limits are not supported for the VID.
2. A zero length list, n = 0 means no limits are currently defined for the specified variable.

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6-3. Stream5 Exception Handling

S5F0 Abort Transaction(S5F0) S,H<->E

Description: Same from as S1F0

S5F1 Alarm Report Send(ARS) S,H<-E,reply

Description: This message reports a change in or presence of an alarm condition. One message will be issued when the alarm is set and one message will be issued when the alarm is cleared. Irrecoverable errors and attention flags may not have a corresponding clear message.

Structure: L,3
1.<ALCD>
Alarm code byte
bit8 = 1 means alarm set
bit8 = 0 means alarm cleared
bits7 - 1 is alarm category code = not used
2.<ALID>
Alarm identification
3.<ALTX>
Alarm text limited to 45 characters

S5F2 Alarm Report Acknowledge(ARA) S,H->E

Description: Acknowledge or error

Structure: <ACKC5>
Acknowledge Code, 1byte
0 = accepted
64 = Invalid ALID
65 = System Error

S5F3 Enable/Disable Alarm Send(EAS) S,H->E,reply

Description: This message will change the state of the enable bit in the equipment. The enable bit determines if the alarm will be sent to the host. Alarms which are not controllable in this way are unaffected by this message.

Structure: L,2
1.<ALED>
Alarm enable/disable code, 1byte
bit8 = 1 means enable alarm
bit8 = 0 means disable alarm
2.<ALID>
Alarm identification

Exception: A zero length item for ALID means all alarms.

S5F4 Enable/Disable Alarm Acknowledge(EAA) S,H<-E

Description: Acknowledge or error.

Structure: <ACKC5>
Acknowledge Code, 1byte
0 = accepted
64 = Invalid ALID
65 = System Error

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S5F5 List Alarm Request(LAR) S,H->E,reply

Description: This message requests the equipment to send binary and analog alarm information to the host.

Structure:
```
<ALID1,...,ALIDn>
  Alarm identification
```

Exception: A zero length item means send all possible alarms regardless of the state of ALED.

S5F6 List Alarm Data(LAD) M,H<-E

Description: This message contains the alarm data known to the equipment. There are "m" alarms in the list.

Structure:
```
L,m
  1.L,3
    1.<ALCD1>
      Alarm code byte
        bit8 = 1 means alarm set
        bit8 = 0 means alarm cleared
        bits7 - 1 is alarm category code = not used
    2.<ALID1>
      Alarm identification
    3.<ALTX1>
      Alarm text limited to 45 characters
  .
  .
  m.L,3
    1.<ALCDm>
    2.<ALIDm>
    3.<ALTXm>
```

Exception: If m = 0, no response can be made. A zero length item returned for ALCDi or ALTXi means that value does not exist.

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6-4. Stream6 Data Collection

S6F0 Abort Transaction(S6F0) S,H<->E

Description: Same from as S1F0

S6F1 Trace Data Send(TDS) S,H<-E,reply

Description: This function sends samples to the host according to the trace setup done by S2F23.
Trace is a time-driven form of equipment status.

Structure: L,4
1.<TRID>
Trace request ID
2.<SMPLN>
Sample number
3.<STIME>
Sample time of day, 16bytes, However this item could be 12bytes or 14bytes.
If 16bytes, the format is YYYYMMDDhhmmsscc,
If 14bytes, the format is YYYYMMDDhhmmss,
If 12bytes, the format is YYMMDDhhmmss.

YYYY	=	Year	0000	to	9999
MM	=	Month	01	to	12
DD	=	Day	01	to	31
hh	=	Hour	00	to	59
mm	=	Minute	00	to	59
ss	=	Second	00	to	59
cc	=	Centisecond	00	to	99

Note: This item format is associated ECV(Time Format) which is Supported as a configurable variable.
Refer to ECV, Time Format.
This is a format requirement only and does not imply either precision.

4.L,n
1.<SV1>
Status variable value
2.<SV2>
.
.
n.<SVn>

Exception: A zero length STIME means no value is given and that the time is to be derived from SMPLN along with knowledge of the request.

S6F2 Trace Data Acknowledge(TDA) S,H->E

Description: Acknowledge or error.

Structure: <ACKC6>
Acknowledge Code, 1byte
0 = accepted and done
65 = System Error

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S6F5 Multi-block Data Send Inquire(MBI) S,H<-E,reply

Description: If the discrete data report S6F11 can involve more than one block, this transaction must precede the transmission.

Structure:
```
L,2
  1.<DATAID>
    Data ID
  2.<DATALENGTH>
    Total bytes to be sent
```

S6F6 Multi-block Grant(MBG) S,H->E

Description: Grant permission to send.

Structure:
```
<GRANT6>
Permission to send, 1byte
  0 = Permission granted
  1 = Busy, try again
  2 = Not interested
  65 = System Error
```

S6F11 Event Report Send(ERS) M,H<-E,reply

Description: The purpose of this message is for the equipment to send a defined, linked and enabled group of reports to the host upon the occurrence of an event(CEID). If S6F11 is Multi-block, it must be preceded by the S6F5/S6F6 Inquire/Grant transaction.

Structure:
```
L,3
  1.<DATAID>
    Data ID
  2.<CEID>
    Collected event ID
  3.L,a
    1.L,2                          report 1
      1.<RPTID1>
        Report ID
      2.L,b
        1.<V1>
          Variable data
        .
        .
        b.<Vb>
    .
    .
    1.L,2                          report a
      1.<RPTIDa>
      2.L,c                        # Vs this report
        1.<V1>
        .
        .
        c.<Vc>
```

Exception: If there are no reports linked to the event a 'null' report is assumed. A zero length list for # of reports means there are no reports linked to the given CEID.

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S6F12 Event Report Acknowlege(ERA) S,H->E

Description: Acknowledge or error

Structure:
```
<ACKC6>
Acknowledge Code, 1byte
  0 = accepted
  65 = System Error
```

S6F15 Event Report Request(ERR) S,H->E,reply

Description: The purpose of this message is for the host to demand a given report group from the equipment.

Structure:
```
<CEID>
Collected event ID
```

S6F16 Event Report Data(ERD) M,H<-E

Description: Equipment sends reports linked to given CEID to host.

Structure: Identical to structure of S6F11. If S6F11 is Multi-block, it must be preceded by the S6F5/S6F6 Inquire/Grant transaction.

```
L,3
  1.<DATAID>
    Data ID
  2.<CEID>
    Collected event ID
  3.L,a
    1.L,2                          report 1
      1.<RPTID1>
        Report ID
      2.L,b
        1.<V1>
          Variable data
        .
        b.<Vb>
    .
    1.L,2                          report a
      1.<RPTIDa>
      2.L,c                  # Vs this report
        1.<V1>
        .
        c.<Vc>
```

If there are no reports linked to the event a 'null' report is assumed. A zero length list for # of reports means there are no reports linked to the given CEID.

Exception: A zero length item means there are no reports linked to the given CEID.

S6F19 Individual Report Request(IRR) S,H->E,reply

Description: The purpose of this message is for host to request a defined report from the equipment.

Structure:
```
<RPTID>
Report ID
```

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S6F20 Individual Report Data(IRD) M,H<-E

Description: Equipment sends variable data defined for the given RPTID to the host.

Structure:
```
L,n
  1.<V1>
    Variable data
  .
  .
  n.<Vn>
```

Exception: A zero length list means RPTID is not defined.

S6F23 Request Spooled Data(RSD) S,H->E,reply

Description: The purpose of this message is for the host to request transmission or deletion of the messages currently spooled by the equipment.

Structure:
```
<RSDC>
Request Spool Data Code
  0 = Transmit Spooled Messages
  1 = Purge Spooled Messages
```

S6F24 Request Spooled Data Acknowledgement Send(RSDAS) S,H<-E

Description: The purpose of this message is to acknowledge the receipt of the Request Spooled Data(S6F23) and to respond with an appropriate acknowledge code.

Structure:
```
<RSDA>
Request Spool Data Acknowledge
  0 = OK
  1 = Denied, busy try later
  2 = Denied, spooled data does not exist
  65 = System Error
```

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6-5. Stream7 Process Program Management

S7F0 Abort Transaction(S7F0) S,H<->E

Description: Same from as S1F0

S7F1 Process Program Load Inquire (PPI) S,H<->E,reply

Description: This message is used to initiate the transfer of a process program(S7F3/S7F4).

Structure: L,2
1.<PPID>
Process program ID.
2.<LENGTH>
Length of the process program in bytes.

S7F2 Process Program Load Grant (PPG) S,H<->E

Description: This message gives permission for the process program to be loaded.

Structure: <PPGNT>
Process program grant status, 1byte
0 = OK
1 = OK (already have)
2 = no space
3 = invalid PPID
4 = busy, try later
5 = will not accept

S7F3 Process Program Send(PPS) M,H<->E,reply

Description: The program is sent.

Structure: L,2
1.<PPID>
Process program ID.
2.<PPBODY>
Process program body

S7F4 Process Program Acknowledge(PPA) S,H<->E

Description: Acknowledge or error.

Structure: <ACKC7>
Acknowledge Code, 1byte
0 = Accepted
1 = Permission not granted
2 = Length error
3 = Matrix overflow
4 = PPID not found
5 = Mode unsupported
6 = Not REMOTE mode
7 = S7F1 not accepted
8 = Operator editing PP
9 = Invalid PPID
10 = Current PPID
11 = Invalid PPBODY(Total length or Item limit error)
>11 = Other error
12 - 63 = reserved

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S7F5 Process Program Request(PPR) S,H<->E,reply

Description: This message is used to request the transfer of a process program.

Structure: <PPID>
Process program ID.

S7F6 Process Program Data(PPD) M,H<->E

Description: This message is used to transfer a process program.

Structure: L,2
1.<PPID>
Process program ID.
2.<PPBODY>
Process program body

Exception: A zero length list means request denied.

Note: The equipment to host transfer of the process program, denoted by the R bit in the header (R = 1), provides the mechanism for the host computer to receive process programs created on the equipment. This allows use of the equipment without having process program generation capabilities on the host.

S7F17 Delete Process Program Send(DPS) S,H->E,reply

Description: This message is used by the host to request the equipment to delete process programs from equipment storage.

Structure: L,n (Number of process programs to be deleted)
1.<PPID1>
Process program ID.
.
.
n.<PPIDn>

Exception: If n = 0, then delete all.

S7F18 Delete Process Program Acknowledge(DPA) S,H<-E

Description: Acknowledge or error.

Structure: <ACKC7>
Acknowledge Code, 1byte
0 = Accepted
1 = Permission not granted
2 = Length error
3 = Matrix overflow
4 = PPID not found
5 = Mode unsupported
6 = Not REMOTE mode
7 = S7F1 not accepted
8 = Operator editing PP
9 = Invalid PPID
10 = Current PPID
11 = Invalid PPBODY(Total length or Item limit error)
>11 = Other error
12 - 63 = reserved

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S7F19 Current EPPD Request(RER)	S,H->E,reply

Description: This message is used to request the transmission of the current equipment process program directory(EPPD). This is a list of all the PPIDs of the process programs stored in the equipment.

Structure: Header only

S7F20 Current EPPD Data(RED)	M,H<-E

Description: This message is used to transmit the current EPPD.

Structure: L,n (number of process programs in the directory)
1.<PPID1>
Process program ID.
.
.
n.<PPIDn>

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6-6. Stream9 System Errors

S9F0 Abort Transaction(S9F0)	S,H<->E

Description: Same from as S1F0

S9F1 Unrecognized Device ID(UDN)	S,H<-E

Description: The device ID in the message block header did not correspond to any known device ID in the node detecting the error.

Structure: <MHEAD>
SECS message block header associated with message block in error

S9F3 Unrecognized Stream Type(USN)	S,H<-E

Description: The equipment does not recognize the stream type in the message block header.

Structure: <MHEAD>
SECS message block header associated with message block in error

S9F5 Unrecognized Function Type(UFN)	S,H<-E

Description: This message indicates that the function in the message ID is not recognized by the receiver.

Structure: <MHEAD>
SECS message block header associated with message block in error

S9F7 Illegal Data(IDN)	S,H<-E

Description: This message indicates that the stream and function were recognized but the associated data format could not be interpreted.

Structure: <MHEAD>
SECS message block header associated with message block in error

S9F9 Transaction Timer Timeout(TTN)	S,H<-E

Description: This message indicates that a transaction(receive) timer has timed out and that he corresponding transaction has been aborted. It is up to the host to respond to this error in an appropriate manner to keep the system operational.

Structure: <SHEAD>
Stored header related to the transaction timer

S9F11 Data Too Long(DLN)	S,H<-E

Description: This message to the host indicates that the equipment has been sent more data that it can handle.

Structure: <MHEAD>
SECS message block header associated with message block in error

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S9F13 Conversation Timeout(CTN)	S,H<-E

Description: Though data should have been received, Host was not able to receive it before appropriate time.
The resource is cleared.

Structure: L,2
1.<MEXP>
The message which you should receive SxxFyy.
2.<EDID>
The data ID which you should receive.

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6-7. Stream10 Terminal Services

S10F0 Abort Transaction(S10F0) S,H<->E

Description: Same from as S1F0

S10F1 Terminal Request(TRN) S,H<-E,reply

Description: A terminal text message to the host.

Structure: L,2
1.<TID>
Terminal number, 1byte
0 = single or main terminal
2.<TEXT>
A single line of characters. (160byte)

S10F2 Terminal Request Acknowledge(TRA) S,H->E

Description: Acknowledge or error

Structure: <ACKC10>
Acknowledge Code, 1byte
0 = Accepted for display
1 = Message will not be displayed
2 = Terminal not available
65 = System Error

S10F3 Terminal Display, Single(VTN) S,H->E,reply

Description: Data to be displayed.

Structure: L,2
1.<TID>
Terminal number, 1byte
0 = single or main terminal
2.<TEXT>
A single line of characters. (160byte)

S10F4 Terminal Display, Single Acknowledge(VTA) S,H<-E

Description: Acknowledge or error

Structure: <ACKC10>
Acknowledge Code, 1byte
0 = Accepted for display
1 = Message will not be displayed
2 = Terminal not available
65 = System Error

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7．Data Item Dictionary

7-1. Item Dictionary

This section defines the data items used in the standard SECS-II messages picked up for TOWA HOST COMMUNICATION SYSTEM.

The description of 'Name', 'Format', 'Description', 'Where' are specified with SEMI E5-0698.

Especially, this section defines about 'Format' and data item length.

ABS **Format:10**

Any binary string

Where Used: S2F25,S2F26

ACKC5 **Format:10**

Acknowledge Code, 1byte
0 = Accepted
65 = System error

Where Used: S5F2,S5F4

ACKC6 **Format:10**

Acknowledge Code, 1byte
0 = Accepted and done
65 = System error

Where Used: S6F2,S6F12

ACKC7 **Format:10**

Acknowledge Code, 1byte
0 = Accepted
1 = Permission not granted
2 = Length error
3 = Matrix overflow
4 = PPID not found
5 = Mode unsupported
6 = Not REMOTE mode
7 = S7F1 not accepted
8 = Operator editing PP
9 = Invalid PPID
10 = Current PPID
11 = Invalid PPBODY(Total length or Item limit error)
>11 = Other error
12 - 63 = reserved

Where Used: S7F4,S7F18

ACKC10 **Format:10**

Acknowledge Code, 1byte
0 = Accepted for display
1 = Message will not be displayed
2 = Terminal not available
65 = System error

Where Used: S10F2,S10F4

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ALCD **Format:10**

Alarm code byte
bit8 = 1 means alarm set
bit8 = 0 means alarm cleared
bits7 - 1 is alarm category code = not used

Where Used: S5F1,S5F6

ALED **Format:10**

Alarm enable/disable code, 1byte
bit8 = 1 means enable alarm
bit8 = 0 means disable alarm

Where Used: S5F3

ALID **Format:52,54**

Alarm identification
(YPM,PMC=Format:54)

Where Used: S5F1,S5F3,S5F5,S5F6

ALTX **Format:20**

Alarm text fixed to 45 characters

Where Used: S5F1,S5F6

CEED **Format:11**

Collection event enable/disable code, 1 byte
FALSE = Disable
TRUE = Enable

Where Used: S2F37

CEID **Format:52**

Collected event ID

Where Used: S2F35,S2F37,S6F11,S6F15,S6F16

COMMACK **Format:10**

Establish Communications Acknowledge Code, 1byte
0 = Accepted
1 = Denied, Try Again
65 = System error

Where Used: S1F14

CPACK **Format:10**

Command Parameter Acknowledge Code, 1byte
1 = Parameter Name(CPNAME) does not exist
2 = Illegal Value specified for CPVAL
3 = Illegal Format specified for CPVAL
>3 = Other equipment-specific error
4 - 63 = reserved

Where Used: S2F42

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CPNAME **Format:20**

See ID Lists 8-4-2. COMMAND PARAMETER.

Where Used: S2F41,S2F42

CPVAL **Format:20, 52**

Command Parameter Value

Where Used: S2F41

DATAID **Format:52**

Data ID

Where Used: S2F33,S2F35,S2F39,S2F45,S6F5,S6F11,S6F16

DATALENGTH **Format:52**

Total bytes to be sent

Where Used: S2F39,S6F5

DRACK **Format:10**

Define Report Acknowledge Code, 1 byte
0 = Accepted
1 = Denied. Insufficient space
2 = Denied. Invalid format
3 = Denied. At least one RPTID already defined
4 = Denied. At least one VID does not exist

Where Used: S2F34

DSPER **Format:20**

Data sample period
hhmmss, 6bytes

Where Used: S2F23

EAC **Format:10**

Equipment acknowledge code, 1byte
0 = Acknowledge
1 = Denied. At least one constant does not exist
2 = Denied. Busy
3 = Denied. At least one constant out of range
65 = System error

Where Used: S2F16

ECDEF **Format:10,11,20,21,3(),4(),5()**

Equipment constant default value

Where Used: S2F30

ECID **Format:52**

Equipment Constant ID

Where Used: S2F13,S2F15,S2F29,S2F30

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ECMAX **Format:10,11,20,21,3(),4(),5()**

Equipment constant maximum value

Where Used: S2F30

ECMIN **Format:10,11,20,21,3(),4(),5()**

Equipment constant minimum value

Where Used: S2F30

ECNAME **Format:20**

Equipment constant name fixed to 6 characters

Where Used: S2F30

ECV **Format:10,11,20,21,3(),4(),5()**

Equipment Constant Value

Where Used: S2F14,S2F15

EDID **Format:10,20,3(),5()**

The data ID which you should receive.
The one of the next is thought about.
MEXP EDID EDID
S9F13 <PPID> A[30],B[30]

Where Used: S9F13

ERACK **Format:10**

Enable/Disable Event Report Acknowledge Code, 1 byte
0 = Accepted
1 = Denied. At least one CEID does not exist

Where Used: S2F38

FCNID **Format:51**

Identification

Where Used: S2F43,S2F44

GRANT **Format:10**

Grant code, 1 byte
0 = positive response, load OK
1 = busy, try again
2 = no space
3 = duplicate name
65 = System error

Where Used: S2F40

GRANT6 **Format:10**

Permission to send, 1byte
0 = Permission granted
1 = Busy, try again
2 = Not interested
65 = System error

Where Used: S6F6

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HCACK **Format:10**

Host Command Parameter Acknowledge Code, 1byte
0 = Acknowledge, command has been performed
1 = Command does not exist
2 = Cannot perform now
3 = At least one parameter is invalid
4 = Acknowledge, command will be performed with completion signaled later by an event.
5 - 63 = Reserved

Where Used: S2F42

LENGTH **Format:54**

Length of the process program in bytes

Where Used: S7F1

LIMITACK **Format:10**

Acknowledgment code for variable limit attribute set, 1byte
1 = LIMITID does not exist.
2 = UPPERDB > LIMITMAX.
3 = LOWERDB < LIMITMIN.
4 = UPPERDB < LOWERDB.
5 = Illegal format specified for UPPERDB or LOWERDB.
6 = ASCII value limit definition for this variable.

Where Used: S2F46

LIMITID **Format:10**

The identifier of a specific limit in the set of limits (as defined by UPPERDB and LOWERDB) for a variable to which the corresponding limit attributes refer, 1byte.

Where Used: S2F45,S2F46,S2F48

LIMITMAX **Format:11,20,3(),4(),5()**

The maximum allowed value for the limit values of a specific variable. The equipment manufacturer should specify this value, which would typically coincide with the maximum value of the variable being monitored. The format must match that of the referenced variable.

Where Used: S2F48

LIMITMIN **Format:11,20,3(),4(),5()**

The minimum allowed value for the limit values of a specific variable. The equipment manufacturer should specify this value, which would typically coincide with the minimum value of the variable being monitored. The format must match that of the referenced variable.

Where Used: S2F48

LOWERDB **Format:11,20,3(),4(),5()**

The variable limit attributes which defines the lower boundary of the deadband of a limit. The value applies to a single limit (LIMITID) for a specified VID. Thus, UPPERDB and LOWERDB as a pair define a limit.

Where Used: S2F45,S2F48

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LRACK **Format:10**

Link Report Acknowledge Code,1byte
0 = Accepted
1 = Denied. Insufficient space
2 = Denied. Invalid format
3 = Denied. At least one CEID link already defined
4 = Denied. At least one CEID does not exist
5 = Denied. At least one RPTID does not exist

Where Used: S2F36

LVACK **Format:10**

Variable limit definition acknowledges code, 1byte. Defines the error with the limit attributes for the referenced VID.
1 = Variable does not exist.
2 = Variable has no limits capability.
3 = Variable repeated in message.
4 = Limit value error as described in LIMITACK.

Where Used: S2F46

MDLN **Format:20**

Equipment Model Type, 6bytes
Same data as returned by S1F2

Where Used: S1F2,S1F13,S1F14

MEXP **Format:20**

The message which you should receive SxxFyy.

Where Used: S9F13

MHEAD **Format:10**

SECS message block header associated with message block in error, 10bytes

Where Used: S9F1,S9F3,S9F5,S9F7,S9F11

OFLACK **Format:10**

Acknowledge code for OFF-LINE request.
0 = OFF-LINE Acknowledge.
65 = System Error

Where Used: S1F16

ONLACK **Format:10**

Acknowledge code for ON-LINE request.
0 = ON-LINE Accepted.
65 = System Error

Where Used: S1F18

PPBODY **Format:10**

Process program body

Where Used: S7F3,S7F6

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PPGNT **Format:10**

Process program grant status, 1byte
0 = OK
1 = OK (already have)
2 = no space
3 = invalid PPID
4 = busy, try later
5 = will not accept
>5 = other error
6 - 63 = reserved

Where Used: S7F2

PPID **Format:20**

Process program ID, fixed to 20bytes.

Where Used: S7F1,S7F3,S7F5,S7F6,S7F17,S7F20

RCMD **Format:20**

See ID Lists 8-4-1. REMOTE COMMAND.

Where Used: S2F41

REPGSZ **Format:52**

Reporting group size

Where Used: S2F23

RPTID **Format:52**

Report ID

Where Used: S2F33,S2F35,S6F11,S6F16,S6F19

RSPACK **Format:10**

Reset Spooling Acknowledge, 1 byte
0 = Acknowledge, spooling setup accepted
1 = Spooling setup rejected.

Where Used: S2F44

RSDA **Format:10**

Request Spool Data Acknowledge, 1 byte
0 = OK
1 = Denied, busy try later
2 = Denied, spooled data does not exist
65 = System Error

Where Used: S6F24

RSDC **Format:51**

Request Spool Data Code
0 = Transmit Spooled Messages
1 = Purge Spooled Messages

Where Used: S6F23

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SHEAD **Format:10**

Stored header related to the transaction timer

Where Used: S9F9

SMPLN **Format:52**

Sample number

Where Used: S6F1

SOFTREV **Format:20**

Software revision code 6bytes

Where Used: S1F2,S1F13,S1F14

STIME **Format:20**

Sample time of day, 16bytes, However this item could be 12bytes or 14bytes.
If 16bytes, the format is YYYYMMDDhhmmsscc,
If 14bytes, the format is YYYYMMDDhhmmss,
If 12bytes, the format is YYMMDDhhmmss.

YYYY	=	Year	0000	to	9999
MM	=	Month	01	to	12
DD	=	Day	01	to	31
hh	=	Hour	00	to	59
mm	=	Minute	00	to	59
ss	=	Second	00	to	59
cc	=	Centisecond	00	to	99

Note: This item format is associated ECV(Time Format) which is Supported as a configurable variable. Refer to ECV, Time Format. This is a format requirement only and does not imply either precision.

Where Used: S6F1

STRACK **Format:10**

Spool Stream Acknowledge

1 = Spooling not allowed for stream(i.e., Stream 1).
2 = Stream unknown.
3 = Unknown function specified for this stream.
4 = Secondary function specified for this stream.

Note: Error codes 1 - 4 shown in order of precedence for reporting. Code 0 not used.

Where Used: S2F44

STRID **Format:51**

Stream Identification

Where Used: S2F43, S2F44

SV **Format:0,10,11,20,21,3(),4(),5()**

Status variable value

Where Used: S1F4, S6F1

SVID **Format:52**

Status variable ID

Where Used: S1F3, S1F11, S1F12, S2F23

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SVNAME **Format:20**

Status Variable Name fixed to 6 characters

Where Used: S1F12

TEXT **Format:20**

A single line of characters fixed to 160 characters

Where Used: S10F1, S10F3

TIAACK **Format:10**

Equipment acknowledgement code, 1 byte
0 = everything correct
1 = too many SVIDs
2 = no more traces allowed
3 = invalid period
65 = System Error

Where Used: S2F24

TIACK **Format:10**

Time Acknowledge Code, 1 byte
0 = OK
1 = Error, not done
65 = System Error

Where Used: S2F24

TID **Format:10**

Terminal number, 1byte
0 = single or main terminal
>0 = additional terminals at the same equipment

Where Used: S10F1, S10F3

TIME **Format:20**

Time of day, 16bytes, However this item could be 12bytes or 14bytes.
If 16bytes, the format is YYYYMMDDhhmmsscc,
If 14bytes, the format is YYYYMMDDhhmmss,
If 12bytes, the format is YYMMDDhhmmss.

YYYY	=	Year	0000	to	9999
MM	=	Month	01	to	12
DD	=	Day	01	to	31
hh	=	Hour	00	to	59
mm	=	Minute	00	to	59
ss	=	Second	00	to	59
cc	=	Centisecond	00	to	99

Note: This item format is associated ECV(Time Format) which is Supported as a configurable variable. Refer to ECV, Time Format. This is a format requirement only and does not imply either precision.

Where Used: S2F18,S2F31

TOTSMP **Format:54**

Total samples to be made

Where Used: S2F23,S6F1

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TRID **Format:52**

Trace request ID

Where Used: S2F23,S6F1

UNITS **Format:20**

Units Identifier, 6bytes

Where Used: S1F12,S2F30,S2F48

UPPERDB **Format:11,20,3(),4(),5()**

A variable limit attributes which defines the upper boundary of the dead band of a limit. The value applies to a single limit(LIMITID) for a specified VID. Thus, UPPERDB and LOWERDB as a pair define a limit.

Where Used: S2F45,F2F48

V **Format:0,10,11,20,21,3(),4(),5()**

Variable data

Where Used: S6F11,S6F16,S6F20

VID **Format:52**

Variable ID

Where Used: S2F33,S2F45,S2F46,S2F47,S2F48

VLAACK **Format:10**

Variable Limit Attribute Acknowledge Code, 1byte
0 = Acknowledge, command will be performed
1 = Limit attribute definition error
2 = Cannot perform now
65 = System Error

Where Used: S2F46

7-2. Item Format Code

Code Octal	Chara	Description
00	L	LIST (length in elements)
10	B	Binary
11	BOOLEAN	Boolean
20	A	ASCII
21	JIS-8	JIS-8
30	I8	8bytes integer (signed)
31	I1	1byte integer (signed)
32	I2	2bytes integer (signed)
34	I4	3bytes integer (signed)
40	F8	8bytes floating point
44	F4	4bytes floating point
50	U8	8bytes integer (unsigned)
51	U1	1bytes integer (unsigned)
52	U2	2bytes integer (unsigned)
54	U4	4bytes integer (unsigned)

※This system do not have Format Code 22.

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